

Geometry — revision of Test 1

Lessons 1–6 · Quarter I · 40 questions

Sketch every question before answering it, and name the fact you used — the angle sum, the property, or the test. A number with no reason earns no method marks.

NAME _____

CLASS _____

DATE _____

Lessons 1–2 · Revision of the Grade 7 course 9

Angles and triangles

- 1

Two angles of a triangle are 42° and 63° . Find the third.
- 2

An isosceles triangle has a base angle of 55° . Find the apex angle.
- 3

An isosceles triangle has an apex angle of 96° . Find each base angle.
- 4

Two interior angles of a triangle are 40° and 65° . Find the exterior angle at the third vertex.
- 5

One acute angle of a right-angled triangle is 37° . Find the other.
- 6

State each angle of an equilateral triangle.
- 7

Two angles lie on a straight line and one is 118° . Find the other.
- 8

Two straight lines cross. One angle is 74° . Find the angle vertically opposite it.
- 9

The angles of a triangle are in the ratio $2 : 3 : 4$. Find them.

Lessons 3–4 · Polygons. Interior and exterior angles 14

Angle sums, regular polygons

- 10

Find the sum of the interior angles of a hexagon.
- 11

Find the sum of the interior angles of a decagon.
- 12

The interior angles of a convex polygon add to 900° . How many sides?
- 13

The interior angles of a convex polygon add to 2160° . How many sides?
- 14

Each exterior angle of a regular polygon is 24° . How many sides?
- 15

Each exterior angle of a regular polygon is 72° . How many sides?
- 16

Find each interior angle of a regular pentagon.
- 17

Find each interior angle of a regular octagon.
- 18

Each interior angle of a regular polygon is 150° . How many sides?
- 19

What do the exterior angles of *any* convex polygon add to?
- 20

A regular polygon has 20 sides. Find each exterior angle.
- 21

Four angles of a pentagon are 100° , 110° , 120° and 95° . Find the fifth.

22 The interior angles of a convex polygon add to 1260° . How many sides?

23 Can a regular polygon have an exterior angle of 50° ? Explain.

Lesson 5 · Parallelogram and its properties 9

Opposite sides and angles, diagonals

24 In parallelogram $ABCD$, $\angle A = 68^\circ$. Find $\angle B$, $\angle C$ and $\angle D$.

28 In parallelogram $ABCD$, $\angle A : \angle B = 2 : 3$. Find both angles.

25 The perimeter of a parallelogram is 42 cm and one side is 12 cm. Find the other side.

29 One angle of a parallelogram is 90° . What can you say about the other three?

26 A parallelogram has $AB = 7$ cm and $BC = 11$ cm. Find its perimeter.

30 In parallelogram $ABCD$ the diagonal BD makes $\angle ABD = 35^\circ$ and $\angle ADB = 60^\circ$. Find $\angle A$.

27 The diagonal AC of parallelogram $ABCD$ is 16 cm and the diagonals meet at O . Find AO .

31 What do two neighbouring angles of a parallelogram add to?

32 In parallelogram $ABCD$, $\angle A = 3\angle B$. Find both angles.

Lesson 6 · Tests for a parallelogram 8

Which test proves it — and which does not

33 In $ABCD$, $AB \parallel CD$ and $AB = CD$. Is it a parallelogram? Name the test.

37 In $ABCD$, $AB \parallel CD$ and $BC = AD$. Must it be a parallelogram?

34 In $ABCD$, $AB = CD$ and $BC = AD$. Is it a parallelogram? Name the test.

38 In $ABCD$, $\angle A = \angle C$ and $\angle B = \angle D$. Is it a parallelogram?

35 In $ABCD$ the diagonals bisect each other. Is it a parallelogram?

39 The diagonals of $ABCD$ meet at O with $AO = OC = 5$ cm and $BO = OD = 7$ cm. Is it a parallelogram? Name the test.

36 In $ABCD$, $AB \parallel CD$ and $BC \parallel AD$. Is it a parallelogram?

40 In $ABCD$, $\angle A = \angle B$ and $\angle C = \angle D$. Must it be a parallelogram?